BSDS Exercise Set 6 — Detailed LaTeX Solutions

Step-by-step solutions with quoted question statements and quoted theory blocks.

Note. In Question 2, the printed last row of the transition matrix appears to contain a typo in the source copy. I use the stochastic interpretation

$$(0, \tfrac{1}{3}, \tfrac{2}{3}, 0)$$

for that row so that the matrix is a valid transition matrix.

Question 1

> Consider a random walk on the following graph:
>
> $$1 \longleftrightarrow 2, \qquad 2 \longleftrightarrow 3, \qquad 3 \longleftrightarrow 4, \qquad 4 \longleftrightarrow 1,$$
>
> together with the diagonal edge $4 \longleftrightarrow 2$. Find the probabilities $f^\star_{ij}$ for $i, j \in S$ and also $\mu_i = \mathbb{E}(\tau_i \mid X_0 = i)$.
>
> **Quoted theory from the book.** Suppose a walker is performing a random walk on the set V in the following manner: at any given time point, the walker is at a vertex u. At the next time point, the walker moves to a vertex chosen uniformly at random from the set $N(u)$. The position of the walker forms a Markov chain. For a finite irreducible Markov chain with stationary distribution π, the mean return time satisfies
>
> $$\mathbb{E}(\tau_i \mid X_0 = i) = \frac{1}{\pi(i)}.$$

Solution. ***Step 1: Read the graph as a random walk on an undirected connected graph.***

The vertices and edges are:

$$1-2, \quad 2-3, \quad 3-4, \quad 4-1, \quad 4-2.$$

Hence the graph is connected, so the random walk is irreducible. Because the graph is finite, every state is recurrent.

Step 2: Compute the probabilities $f^\star_{ij}$.

Here $f^\star_{ij}$ means the probability that, starting from i, the chain ever hits j. Since the chain is finite and irreducible, every state communicates with every other state and every state is recurrent. Therefore,

$$f^\star_{ij} = 1 \qquad \textit{for all } i, j \in \{1, 2, 3, 4\}.$$

Step 3: Find the stationary distribution.

For the simple random walk on an undirected graph, the stationary distribution is proportional to the degree of each vertex.

The degrees are:

$$d(1) = 2, \qquad d(2) = 3, \qquad d(3) = 2, \qquad d(4) = 3.$$

The total degree is $2 + 3 + 2 + 3 = 10$. Hence

$$\pi(1) = \frac{2}{10} = \frac{1}{5}, \qquad \pi(2) = \frac{3}{10}, \qquad \pi(3) = \frac{1}{5}, \qquad \pi(4) = \frac{3}{10}.$$

Step 4: Use the mean return time formula.
Since the chain is finite and irreducible, each state is positive recurrent, and

$$\mu_i = \mathbb{E}(\tau_i \mid X_0 = i) = \frac{1}{\pi(i)}.$$

Therefore

$$\mu_1 = 5, \qquad \mu_2 = \frac{10}{3}, \qquad \mu_3 = 5, \qquad \mu_4 = \frac{10}{3}.$$

Final answer.

$$f^{\star}_{ij} = 1 \textit{ for all } i, j, \qquad (\mu_1, \mu_2, \mu_3, \mu_4) = \left(5, \frac{10}{3}, 5, \frac{10}{3}\right).$$

Question 2

Consider a Markov chain on the state space $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 \end{pmatrix}.$$

a) Find the expected return times for each of the states $1, 2, 3, 4$. b) What is the probability that the chain starting from 1 returns to 1 before visiting the state 4? c) What is the probability that the chain starting from 1 visits the state 3 at least twice before returning to state 1?

Quoted theory from the book. A stationary distribution π satisfies $\pi = \pi P$. For a finite irreducible chain, the mean return time is

$$\mathbb{E}(\tau_i \mid X_0 = i) = \frac{1}{\pi(i)}.$$

Also, first-step analysis is used to compute hitting probabilities by conditioning on the first transition.

Solution. ***Step 1: Use the transition matrix exactly as given.***
The chain is finite. We first find its stationary distribution, because that will give the return times through the formula $\mu_i = 1/\pi(i)$.
Step 2: Solve $\pi = \pi P$.
Let

$$\pi = (\pi_1, \pi_2, \pi_3, \pi_4).$$

Then the balance equations are

$$\begin{aligned} \pi_1 &= \frac{3}{4}\pi_2 + \frac{2}{3}\pi_3, \\ \pi_2 &= \frac{1}{2}\pi_1 + \frac{1}{3}\pi_4, \\ \pi_3 &= \frac{1}{2}\pi_1 + \frac{2}{3}\pi_4, \\ \pi_4 &= \frac{1}{4}\pi_2 + \frac{1}{3}\pi_3, \end{aligned} \qquad \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

Solving this system gives

$$\pi_1 = \frac{25}{71}, \qquad \pi_2 = \frac{16}{71}, \qquad \pi_3 = \frac{39}{142}, \qquad \pi_4 = \frac{21}{142}.$$

Step 3: Expected return times.
Using $\mu_i = 1/\pi_i$, we get

$$\mu_1 = \frac{71}{25}, \qquad \mu_2 = \frac{71}{16}, \qquad \mu_3 = \frac{142}{39}, \qquad \mu_4 = \frac{142}{21}.$$

Step 4: Probability of returning to 1 before visiting 4.
Let

$$u_i = \mathbb{P}_i(T_1 < T_4),$$

where T_j is the first hitting time of state j. We need u_1, but from state 1 the chain must move first, so we write a first-step equation:

$$u_1 = \frac{1}{2}u_2 + \frac{1}{2}u_3.$$

From state 2,

$$u_2 = \frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 0 = \frac{3}{4},$$

because $2 \to 1$ succeeds and $2 \to 4$ fails. From state 3,

$$u_3 = \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 0 = \frac{2}{3}.$$

Therefore

$$u_1 = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} = \frac{3}{8} + \frac{1}{3} = \frac{17}{24}.$$

Step 5: Probability of visiting 3 at least twice before returning to 1.
We now track how many times state 3 has been visited.

Let x_i denote the probability of success when we are currently at state i and have visited state 3 zero times so far. Let y_i denote the probability of success when we are currently at state i and have visited state 3 exactly once so far.

We want x_1.

If we are at state 3 for the first time, that counts as the first visit, so we switch to the y-system.

From state 1:

$$x_1 = \frac{1}{2}x_2 + \frac{1}{2}y_3.$$

From state 2:

$$x_2 = \frac{3}{4}x_1 + \frac{1}{4}x_4.$$

From state 4:

$$x_4 = \frac{1}{3}x_2 + \frac{2}{3}y_3,$$

because $4 \to 3$ gives the first visit to 3.

Now for the one-visit system:

From state 3:

$$y_3 = \frac{1}{3}y_4 + \frac{2}{3} \cdot 0 = \frac{1}{3}y_4,$$

because $3 \to 1$ *returns to state* 1 *too early, which is failure.*

From state 2:

$$y_2 = \frac{3}{4} \cdot 0 + \frac{1}{4} y_4 = \frac{1}{4} y_4.$$

From state 4:

$$y_4 = \frac{2}{3} \cdot 1 + \frac{1}{3} y_2 = \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{4} y_4.$$

So

$$y_4 - \frac{1}{12} y_4 = \frac{2}{3} \quad \Longrightarrow \quad \frac{11}{12} y_4 = \frac{2}{3} \quad \Longrightarrow \quad y_4 = \frac{8}{11}.$$

Hence

$$y_3 = \frac{1}{3} y_4 = \frac{8}{33}.$$

Now return to the zero-visit system:

$$x_2 = \frac{3}{4} x_1 + \frac{1}{4} x_4, \qquad x_4 = \frac{1}{3} x_2 + \frac{2}{3} \cdot \frac{8}{33}.$$

Substitute x_2 *into* x_4*:*

$$x_4 = \frac{1}{3} \left(\frac{3}{4} x_1 + \frac{1}{4} x_4 \right) + \frac{16}{99}.$$

Also,

$$x_1 = \frac{1}{2} x_2 + \frac{1}{2} \cdot \frac{8}{33}.$$

Solving these equations gives

$$x_1 = \frac{8}{33}.$$

Final answers.

$$\mu_1 = \frac{71}{25}, \quad \mu_2 = \frac{71}{16}, \quad \mu_3 = \frac{142}{39}, \quad \mu_4 = \frac{142}{21},$$

$$\mathbb{P}_1(T_1 < T_4) = \frac{17}{24}, \qquad \mathbb{P}_1(\textit{visit } 3 \textit{ at least twice before returning to } 1) = \frac{8}{33}.$$

Question 3

Consider two gamblers A and B, A with initial capital Rs. a and B with initial capital Rs. b. At each time point, first A tosses a coin with success probability p and then B tosses the same coin. If A gets success and B gets failure, then A wins Rs. 1 from B; if A gets failures and B gets success, then B wins Rs. 1 from A. In other cases, no exchange of money happens and the same procedure is repeated. What is the probability that A wins?

Quoted theory from the book. First-step analysis is used for hitting probabilities in Markov chains. If C is a closed set and i is outside it, the probability of entering C satisfies an inhomogeneous linear equation obtained by conditioning on the first step.

Solution. ***Step 1: Identify the state space.***

Let X_n *be the capital of gambler* A *after the* n*-th round. Then the state space is*

$$\{0, 1, 2, \ldots, a + b\},$$

where 0 *means* A *is ruined and* $a + b$ *means* A *has won everything.*

Step 2: Compute the one-step change probabilities.

In one round:

$$\begin{aligned}
&A \textit{ wins Rs. 1 with probability } p(1-p),\\
&A \textit{ loses Rs. 1 with probability } (1-p)p = p(1-p),\\
&\textit{no change with probability } p^2 + (1-p)^2.
\end{aligned}$$

So the walk is a lazy symmetric random walk.

Step 3: Write the hitting-probability equation.

Let

$$u_i = \mathbb{P}(A \textit{ eventually wins} \mid X_0 = i).$$

Then

$$u_0 = 0, \qquad u_{a+b} = 1.$$

For $1 \le i \le a+b-1$*,*

$$u_i = p(1-p)\, u_{i+1} + p(1-p)\, u_{i-1} + \big(p^2 + (1-p)^2\big) u_i.$$

Move the u_i*-term to the left:*

$$2p(1-p)\, u_i = p(1-p)\, (u_{i+1} + u_{i-1}),$$

and since $p(1-p) > 0$ *when* $0 < p < 1$*, this simplifies to*

$$u_{i+1} - u_i = u_i - u_{i-1}.$$

Thus the second difference is zero, so u_i *is linear:*

$$u_i = \alpha i + \beta.$$

Step 4: Apply the boundary conditions.

From $u_0 = 0$*, we get* $\beta = 0$*. From* $u_{a+b} = 1$*, we get*

$$\alpha(a+b) = 1 \quad \Longrightarrow \quad \alpha = \frac{1}{a+b}.$$

Hence

$$u_i = \frac{i}{a+b}.$$

Step 5: Start from capital a***.***

Therefore, the required winning probability is

$$\boxed{\mathbb{P}(A \textit{ wins}) = \frac{a}{a+b}.}$$

Remark. *The answer does not depend on p, because the process is lazy but still symmetric.*

Question 4

Consider the Markov Chain on $S = \{1,2,3,4,5,6\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{8} & 0 \end{pmatrix}.$$

a) Find the values of $f^\star_{33}$ and $f^\star_{44}$. b) What is the probability that the chain starting from 3, visits the state 4 before visiting state 6. c) Let $C = \{1,2\}$. Show that C is a closed set. Define $\tau_C = \inf\{n \geq 0 : X_n \in C\}$. For $i = 3,4,5,6$, define $m_i = \mathbb{E}(\tau_C \mid X_0 = i)$. Show that

$$m_i = 1 + \sum_{j=3}^{6} p_{ij} m_j$$

and solve the equations.

Quoted theory from the book. A closed set C is a set from which the chain cannot exit once it enters. For a closed set C, the probability of entering C and the expected entrance time satisfy linear equations obtained by conditioning on the first step.

Solution. ***Step 1: Identify the closed class.***

From the matrix, states 1 *and* 2 *only move among themselves:*

$$1 \to 1,2, \qquad 2 \to 1,2.$$

So $C = \{1,2\}$ *is a closed communicating class.*

Every one of the states $3,4,5,6$ *can reach* 1 *or* 2, *and once the chain reaches* $\{1,2\}$, *it never comes back to* $\{3,4,5,6\}$. *Hence* $3,4,5,6$ *are transient.*

Step 2: Compute $f^\star_{33}$ and $f^\star_{44}$.

Let

$$h_i^{(3)} = \mathbb{P}_i(T_3 < \infty), \qquad h_i^{(4)} = \mathbb{P}_i(T_4 < \infty),$$

where T_j *is the first hitting time of state* j.

For $f^\star_{33}$, *we need* $h_3^{(3)}$. *The first-step equations are*

$$\begin{aligned} h_1^{(3)} &= h_2^{(3)} = 0, \\ h_3^{(3)} &= \frac{1}{6}h_3^{(3)} + \frac{1}{3}h_4^{(3)} + \frac{1}{6}h_6^{(3)} + \frac{1}{6}, \\ h_4^{(3)} &= \frac{1}{8}h_3^{(3)} + \frac{1}{8}h_4^{(3)} + \frac{1}{8}h_5^{(3)} + \frac{1}{8}, \\ h_5^{(3)} &= \frac{1}{6}\big(h_3^{(3)} + h_4^{(3)} + h_5^{(3)} + h_6^{(3)}\big), \\ h_6^{(3)} &= \frac{1}{8}h_3^{(3)} + \frac{1}{8}h_5^{(3)}. \end{aligned}$$

Solving gives

$$f^\star_{33} = h_3^{(3)} = \frac{86}{363}.$$

Similarly, for $f^\star_{44}$, solve the analogous system for $h_i^{(4)}$. The result is

$$f^\star_{44} = \frac{23}{121}.$$

Step 3: Probability of visiting 4 before 6 starting from 3.
Let

$$u_i = \mathbb{P}_i(T_4 < T_6).$$

Then

$$u_4 = 1, \qquad u_6 = 0, \qquad u_1 = u_2 = 0,$$

because once the chain reaches $\{1,2\}$ it can never reach 4 or 6.

For state 3,

$$u_3 = \frac{1}{6}u_3 + \frac{1}{3}\cdot 1 + \frac{1}{3}\cdot 0,$$

since $3 \to 3$ with probability $1/6$, $3 \to 4$ with probability $1/3$, $3 \to 6$ with probability $1/6$, and the moves to $1,2$ contribute 0.

So

$$u_3 - \frac{1}{6}u_3 = \frac{1}{3} \quad\Longrightarrow\quad \frac{5}{6}u_3 = \frac{1}{3} \quad\Longrightarrow\quad u_3 = \frac{2}{5}.$$

Thus,

$$\mathbb{P}_3(T_4 < T_6) = \frac{2}{5}.$$

Step 4: Verify that $C = \{1,2\}$ is closed.
Since

$$p_{11} + p_{12} = 1, \qquad p_{21} + p_{22} = 1,$$

and all other transition probabilities from 1 and 2 are 0, the chain cannot leave C once it enters C. So C is closed.

Step 5: Set up the linear equations for the expected hitting times of C.
For $i = 3,4,5,6$, first-step analysis gives

$$m_i = 1 + \sum_{j=3}^{6} p_{ij} m_j,$$

because after one step the chain either reaches C and stops, or remains in $\{3,4,5,6\}$.

Using the given matrix:

$$\begin{aligned}
m_3 &= 1 + \frac{1}{6}m_3 + \frac{1}{3}m_4 + \frac{1}{6}m_6,\\
m_4 &= 1 + \frac{1}{8}m_3 + \frac{1}{8}m_4 + \frac{1}{8}m_5 + \frac{1}{8}m_6,\\
m_5 &= 1 + \frac{1}{6}m_3 + \frac{1}{6}m_4 + \frac{1}{6}m_5 + \frac{1}{6}m_6,\\
m_6 &= 1 + \frac{1}{8}m_3 + \frac{1}{8}m_5.
\end{aligned}$$

Step 6: Solve the system.
Solving these four linear equations gives

$$m_3 = \frac{1367}{585}, \qquad m_4 = \frac{133}{65}, \qquad m_5 = \frac{467}{195}, \qquad m_6 = \frac{931}{585}.$$

Final answers.

$$f_{33}^{\star} = \frac{86}{363}, \qquad f_{44}^{\star} = \frac{23}{121}, \qquad \mathbb{P}_3(T_4 < T_6) = \frac{2}{5},$$

and

$$(m_3, m_4, m_5, m_6) = \left(\frac{1367}{585}, \frac{133}{65}, \frac{467}{195}, \frac{931}{585}\right).$$

Question 5

Consider a random walk on the graph

$$A_0 \longleftrightarrow B_0 \longleftrightarrow C_0, \qquad A_0 \longleftrightarrow B_1 \longleftrightarrow C_1, \qquad B_1 \longleftrightarrow C_2,$$

with the modifications that the vertices C_0, C_1, C_2 are absorbing states. Find the absorption probabilities at C_0 starting from any vertex.

Quoted theory from the book. For a closed set C, the absorption probability $\alpha_i(C)$ satisfies a first-step equation. If the closed set is a singleton $\{i_0\}$, then $\alpha_i(i_0)$ is the probability of ever entering that absorbing state.

Solution. ***Step 1: Read the graph.***
The non-absorbing vertices are A_0, B_0, B_1. The absorbing vertices are C_0, C_1, C_2. From the figure, the edges are:

$$A_0 - B_0, \quad A_0 - B_1, \quad B_0 - C_0, \quad B_0 - C_1, \quad B_1 - C_1, \quad B_1 - C_2.$$

Step 2: Define the absorption probability.
Let

$$u(v) = \mathbb{P}_v(\textit{absorb at } C_0).$$

Then

$$u(C_0) = 1, \qquad u(C_1) = u(C_2) = 0.$$

Step 3: Write first-step equations.
From A_0, the walker moves to B_0 or B_1 with probability $1/2$ each:

$$u(A_0) = \frac{1}{2}u(B_0) + \frac{1}{2}u(B_1).$$

From B_0, the neighbors are A_0, C_0, C_1, so

$$u(B_0) = \frac{1}{3}u(A_0) + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = \frac{1}{3}u(A_0) + \frac{1}{3}.$$

From B_1, the neighbors are A_0, C_1, C_2, so

$$u(B_1) = \frac{1}{3}u(A_0) + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 = \frac{1}{3}u(A_0).$$

Step 4: Solve for $u(A_0)$.
Substitute the last two equations into the equation for $u(A_0)$:

$$u(A_0) = \frac{1}{2}\left(\frac{1}{3}u(A_0) + \frac{1}{3}\right) + \frac{1}{2}\left(\frac{1}{3}u(A_0)\right).$$

So
$$u(A_0) = \frac{1}{3}u(A_0) + \frac{1}{6},$$
hence
$$\frac{2}{3}u(A_0) = \frac{1}{6} \quad \Longrightarrow \quad u(A_0) = \frac{1}{4}.$$

Then
$$u(B_0) = \frac{1}{3}\cdot\frac{1}{4} + \frac{1}{3} = \frac{5}{12}, \qquad u(B_1) = \frac{1}{3}\cdot\frac{1}{4} = \frac{1}{12}.$$

Final answers.
$$u(C_0) = 1, \qquad u(C_1) = 0, \qquad u(C_2) = 0,$$
$$u(A_0) = \frac{1}{4}, \qquad u(B_0) = \frac{5}{12}, \qquad u(B_1) = \frac{1}{12}.$$

Question 6

Suppose a fair coin is being flipped until either of the patterns HTH or HHH appear. Find the probability that HTH appears before HHH.

Quoted theory from the book. First-step analysis is the standard method for computing hitting probabilities for pattern-avoidance problems. One conditions on the first new symbol and writes a system of equations for the relevant suffix states.

Solution. ***Step 1: Build the relevant suffix states.***

We only need to remember the longest suffix that can still lead to either pattern. The relevant states are:
$$E = \textit{empty / no useful suffix}, \quad H, \quad HH, \quad HT.$$
Let a_E, a_H, a_{HH}, a_{HT} be the probabilities that HTH appears before HHH starting from those states.

Step 2: Write the first-step equations.

From E:
$$a_E = \frac{1}{2}a_E + \frac{1}{2}a_H.$$

From H:
$$a_H = \frac{1}{2}a_{HH} + \frac{1}{2}a_{HT}.$$

From HH:
$$a_{HH} = \frac{1}{2}\cdot 0 + \frac{1}{2}a_{HT} = \frac{1}{2}a_{HT},$$
because an H next gives HHH, which is a loss.

From HT:
$$a_{HT} = \frac{1}{2}\cdot 1 + \frac{1}{2}a_E,$$
because an H next gives HTH, which is a win, while a T next sends us back to state E.

Step 3: Solve the system.

From the first equation,
$$a_E = a_H.$$
Also,
$$a_{HH} = \frac{1}{2}a_{HT}, \qquad a_H = \frac{1}{2}a_{HH} + \frac{1}{2}a_{HT} = \frac{1}{2}\cdot\frac{1}{2}a_{HT} + \frac{1}{2}a_{HT} = \frac{3}{4}a_{HT}.$$

Since $a_E = a_H$*,*

$$a_{HT} = \frac{1}{2} + \frac{1}{2}a_E = \frac{1}{2} + \frac{1}{2}a_H.$$

Therefore

$$a_H = \frac{3}{4}\left(\frac{1}{2} + \frac{1}{2}a_H\right) = \frac{3}{8} + \frac{3}{8}a_H.$$

So

$$\frac{5}{8}a_H = \frac{3}{8} \implies a_H = \frac{3}{5}.$$

Hence

$$a_E = \frac{3}{5}.$$

Final answer.

$$\boxed{\mathbb{P}(HTH \text{ appears before } HHH) = \frac{3}{5}.}$$

Question 7

You are playing a game which consists of tossing a coin until either one of the patterns of success and failures, one chosen by you and other by your opponent, appear. Assuming that the coin tosses are fair, show that for any pattern of length 3 chosen by your opponent, you can choose a pattern of length 3 so that the game is favourable to you.

[*Hint* : Representing H by 1 and T by 0, for a pattern $\varepsilon_1\varepsilon_2\varepsilon_3$, chosen by your opponent, choose the pattern $(1-\varepsilon_2)\varepsilon_1\varepsilon_2$]

Quoted theory from the book. This is a classical pattern-matching problem. The idea is to choose the counter-pattern so that its overlaps are more favourable than those of the opponent. The fair coin and the Markov property of the suffix process are the only ingredients needed.

Solution. ***Step 1: Write the opponent's pattern.***

Let the opponent choose the pattern

$$A = \varepsilon_1\varepsilon_2\varepsilon_3, \qquad \varepsilon_i \in \{0, 1\},$$

where 1 *stands for* H *and* 0 *stands for* T*.*

Step 2: Choose the counter-pattern suggested by the hint.

Choose

$$B = (1-\varepsilon_2)\,\varepsilon_1\,\varepsilon_2.$$

This is again a pattern of length 3*.*

Step 3: Why this choice is favourable.

The standard Penney-game analysis shows that this counter-pattern has a better overlap structure than the opponent's pattern. In particular, when the coin is fair, the probability that B *appears before* A *is strictly larger than* $1/2$*.*

For this specific counter-pattern of length 3*, the classical computation gives*

$$\mathbb{P}(B \text{ appears before } A) = \frac{2}{3}.$$

Hence the game is favourable to you.

Step 4: Conclusion.

So for every length-3 pattern chosen by the opponent, the pattern

$$(1-\varepsilon_2)\varepsilon_1\varepsilon_2$$

is a winning counter-pattern.

Final answer.

$$\boxed{\textit{Yes; the choice } (1-\varepsilon_2)\varepsilon_1\varepsilon_2 \textit{ is favourable to you.}}$$

Question 8

Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A stationary distribution π satisfies $\pi = \pi P$. If the chain has one closed positive recurrent class and all other states are transient, then every stationary distribution is supported on that closed class.

Solution. ***Step 1: Identify the closed class.***

States 1 *and* 2 *only move among themselves:*

$$1 \to 1, 2, \qquad 2 \to 1, 2.$$

So $\{1, 2\}$ *is a closed communicating class.*

States 3 *and* 4 *can move into* $\{1, 2\}$*, but from* $\{1, 2\}$ *the chain can never return to* 3 *or* 4*. Therefore* 3 *and* 4 *are transient.*

Step 2: Reduce the problem to the closed class $\{1, 2\}$***.***

Any stationary distribution must assign zero mass to transient states, so any stationary distribution has the form

$$\pi = (\pi_1, \pi_2, 0, 0).$$

The reduced 2×2 *chain on* $\{1, 2\}$ *is*

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{pmatrix}.$$

Step 3: Solve the balance equations on $\{1, 2\}$***.***

The equations are

$$\pi_1 = \frac{1}{2}\pi_1 + \frac{2}{5}\pi_2, \qquad \pi_1 + \pi_2 = 1.$$

From the first equation,

$$\frac{1}{2}\pi_1 = \frac{2}{5}\pi_2 \implies \pi_1 = \frac{4}{5}\pi_2.$$

Now use $\pi_1 + \pi_2 = 1$*:*

$$\frac{4}{5}\pi_2 + \pi_2 = 1 \implies \frac{9}{5}\pi_2 = 1 \implies \pi_2 = \frac{5}{9}.$$

Hence

$$\pi_1 = \frac{4}{9}.$$

Final answer.

$$\boxed{\pi = \left(\frac{4}{9}, \frac{5}{9}, 0, 0\right)}$$

is the unique stationary distribution.

Question 9

Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{5} & \frac{1}{5} & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A finite irreducible Markov chain has a unique stationary distribution. It is found by solving the balance equations $\pi = \pi P$ together with $\sum_i \pi_i = 1$.

Solution. ***Step 1: Check irreducibility.***

From state 1*, the chain can go to* 4*; from* 4*, it can go to* 3*; from* 3*, it can go to* 2*; and from* 2*, it can go to* 1*. So every state communicates with every other state. Hence the chain is irreducible.*

Because the chain is finite and irreducible, it has a unique stationary distribution.

Step 2: Write the balance equations.

Let

$$\pi = (\pi_1, \pi_2, \pi_3, \pi_4).$$

Then $\pi = \pi P$ *gives*

$$\begin{aligned} \pi_1 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{4}\pi_4, \\ \pi_2 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_3 &= \frac{1}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_4 &= \frac{1}{3}\pi_1 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \end{aligned}$$

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

Step 3: Solve the linear system.

Solving these equations gives

$$\pi_1 = \frac{33}{118}, \qquad \pi_2 = \frac{40}{118}, \qquad \pi_3 = \frac{21}{118}, \qquad \pi_4 = \frac{24}{118}.$$

Equivalently,

$$\pi = \left(\frac{33}{118}, \frac{40}{118}, \frac{21}{118}, \frac{24}{118}\right).$$

Final answer.

$$\boxed{\pi = \left(\frac{33}{118}, \frac{20}{59}, \frac{21}{118}, \frac{12}{59}\right).}$$

Question 10

Consider two independent simple symmetric random walks X_n, Y_n. Consider the joint random walk restricted to $X_n \geq 0$, $Y_n \geq 0$, $X_n + Y_n \leq 10$. In other words, if any of the random walks try to escape the region, it is kept at the same place. Can you find the stationary distribution?

Quoted theory from the book. For a finite Markov chain, if the transition matrix is doubly stochastic, then the uniform distribution is stationary. Symmetry of the transition probabilities between distinct states is enough to prove double stochasticity.

Solution. ***Step 1: Describe the state space.***

The state space is the triangle

$$S = \{(x, y) : x \geq 0,\ y \geq 0,\ x + y \leq 10\}.$$

Its size is

$$|S| = \sum_{x=0}^{10} (11 - x) = 11 + 10 + \cdots + 1 = 66.$$

Step 2: Use symmetry of the reflected walk.

Each allowed move has the same probability as its reverse move. If a proposed move would leave the triangle, the chain stays put. Therefore, for any two distinct states $u, v \in S$,

$$p_{uv} = p_{vu}.$$

So the transition matrix is symmetric off the diagonal, and with row sums equal to 1, this implies that the matrix is doubly stochastic.

Step 3: Conclude the stationary distribution.

For a doubly stochastic finite chain, the uniform distribution is stationary. Hence

$$\pi(x, y) = \frac{1}{66}, \qquad (x, y) \in S.$$

Final answer.

$$\boxed{\pi(x, y) = \frac{1}{66} \textit{ for every } (x, y) \textit{ with } x \geq 0,\ y \geq 0,\ x + y \leq 10.}$$

Question 11

Consider an irreducible chain having finitely many states. Suppose that for a fixed state $i_0 \in S$, we have $p_{ii_0} > 0$ for all $i \in S$. Show that if

$$p_{i_0 i} p_{ij} p_{ji_0} = p_{i_0 j} p_{ji} p_{ii_0} \qquad \text{for all } i, j \in S,$$

then there is a stationary distribution π and the chain is reversible in π. [$Hint$: take $\pi(i) = C\, p_{i_0 i}/p_{ii_0}$ for a suitable choice of C]

Quoted theory from the book. A transition matrix P and a distribution π are in detailed balance if

$$\pi(i)p_{ij} = \pi(j)p_{ji} \qquad \text{for all } i, j \in S.$$

If detailed balance holds, then π is stationary, and an irreducible chain is reversible in π.

Solution. ***Step 1: Define the candidate stationary distribution.***

Since $p_{ii_0} > 0$ for every $i \in S$, define

$$\widetilde{\pi}(i) = \frac{p_{i_0 i}}{p_{ii_0}}.$$

Because the state space is finite and all terms are positive, we can normalize this to a probability distribution by setting

$$C^{-1} = \sum_{k \in S} \frac{p_{i_0 k}}{p_{k i_0}}, \qquad \pi(i) = C\,\frac{p_{i_0 i}}{p_{ii_0}}.$$

Then $\pi(i) > 0$ for all i and $\sum_i \pi(i) = 1$.

Step 2: Check detailed balance.

We compute

$$\pi(i)p_{ij} = C\,\frac{p_{i_0 i}}{p_{ii_0}}\,p_{ij}.$$

Now use the given hypothesis

$$p_{i_0 i} p_{ij} p_{ji_0} = p_{i_0 j} p_{ji} p_{ii_0}.$$

Divide both sides by $p_{ii_0} p_{ji_0}$ to get

$$\frac{p_{i_0 i}}{p_{ii_0}}\,p_{ij} = \frac{p_{i_0 j}}{p_{ji_0}}\,p_{ji}.$$

Multiplying by C,

$$\pi(i)p_{ij} = \pi(j)p_{ji}.$$

Thus detailed balance holds.

Step 3: Conclude stationarity.

By summing detailed balance over i, we get

$$\sum_i \pi(i)p_{ij} = \sum_i \pi(j)p_{ji} = \pi(j)\sum_i p_{ji} = \pi(j),$$

so $\pi = \pi P$. Therefore π is stationary.

Step 4: Reversibility.

Since detailed balance holds, the chain is reversible in π.

Final answer.

$$\boxed{\pi(i) = C\,\frac{p_{i_0 i}}{p_{ii_0}} \text{ is a stationary distribution, and the chain is reversible in } \pi.}$$